ECE1508 Course Project

Point Cloud Classification Featuring PointNet & PointNet++

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Dec 8, 2025

Introduction

Background

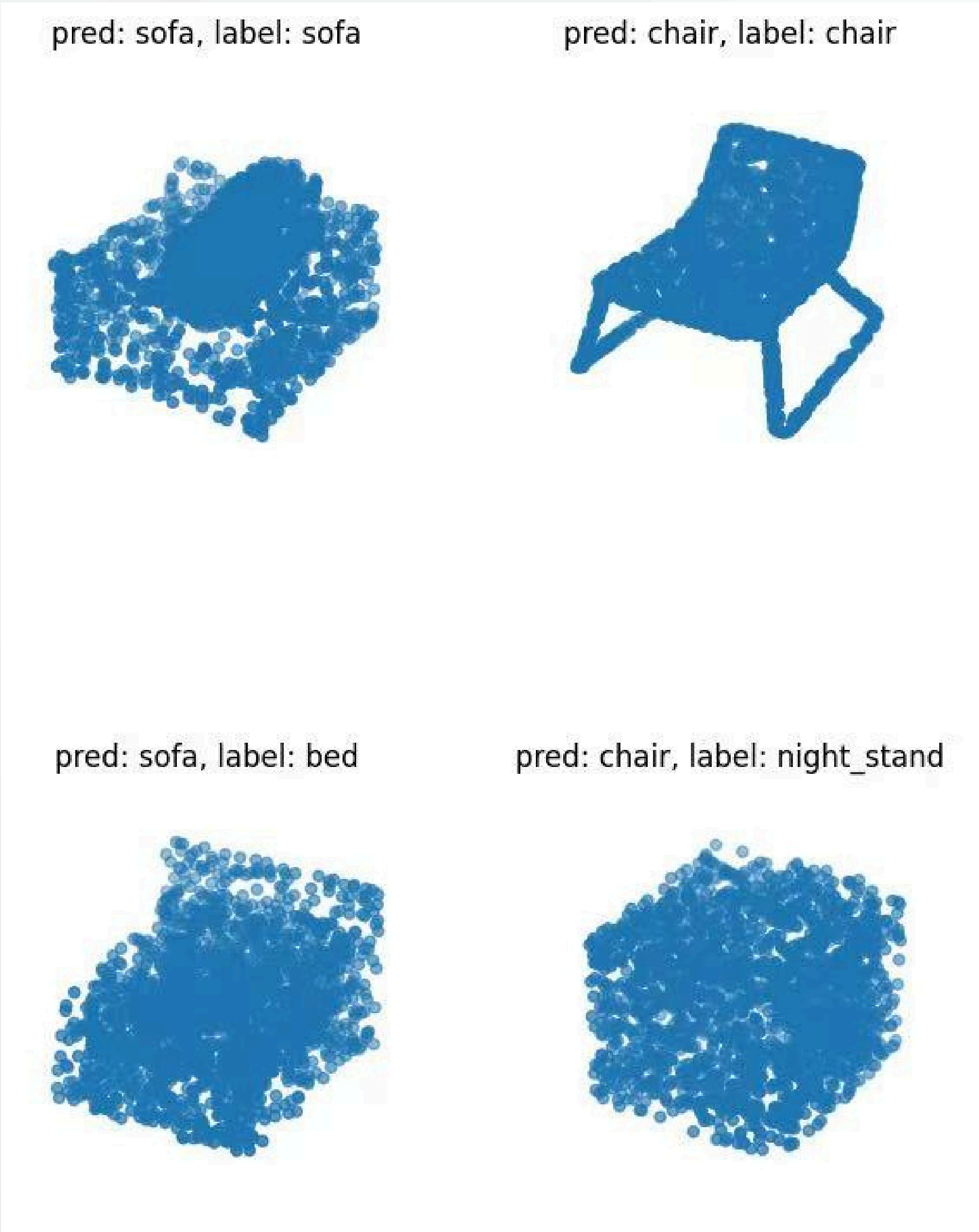
- 3D point clouds are widely used to represent real-world objects (e.g., CAD models, LiDAR scans)
- Their **unordered and irregular** structure makes classification challenging
- PointNet and PointNet++ are specifically designed for such data using **permutation-invariant pooling** and **hierarchical local feature learning.**

Problem Formulation

- Given a point cloud from the ModelNet40 dataset, compare performance of PointNet and PointNet++ (with **data augmentation** and **feature engineering**) to predict its object category.

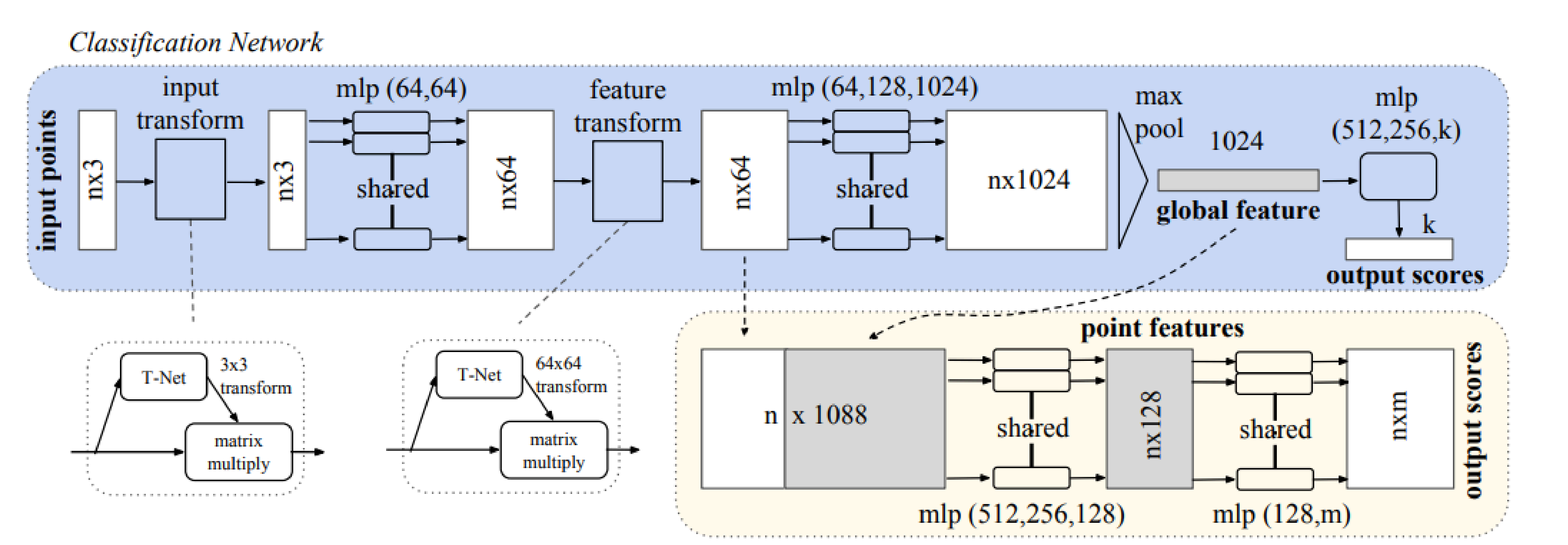
Dataset

- ModelNet40
- ~12k CAD models, 40 categories, ~1024 points/model.

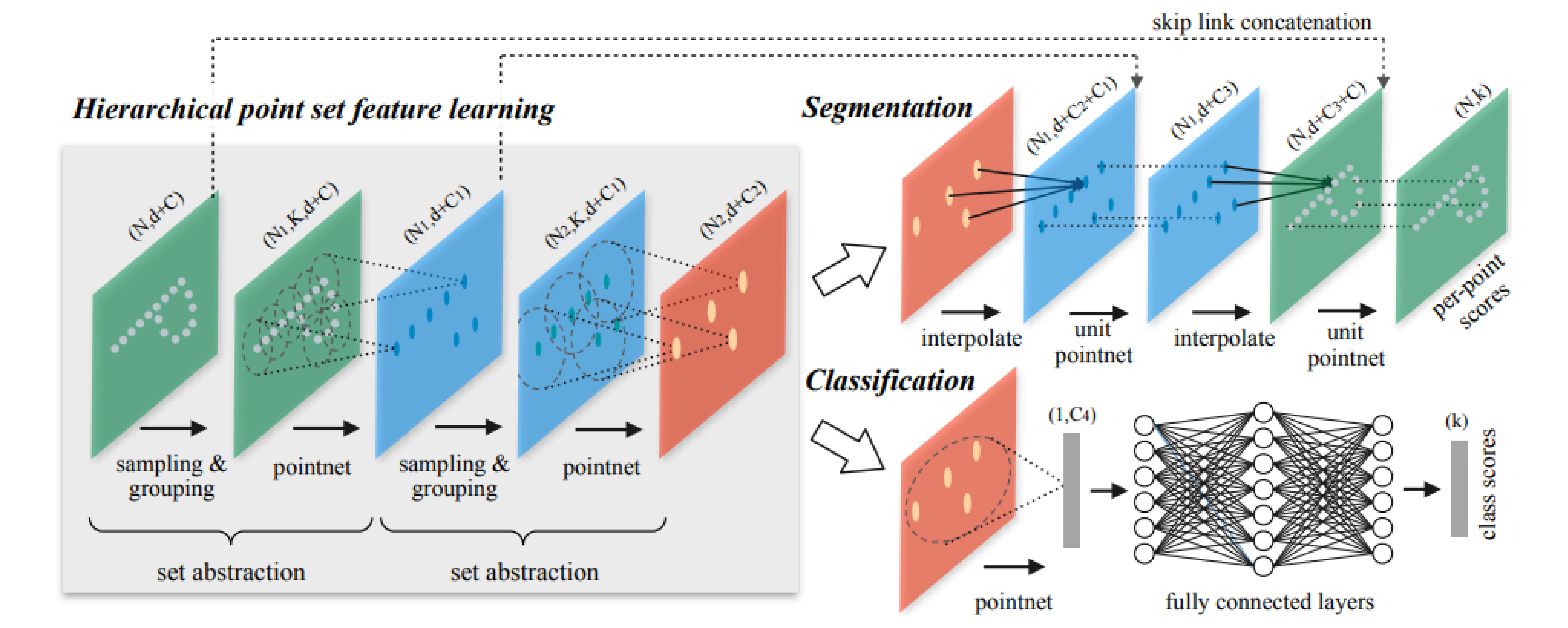


Model Architecture

PointNet



PointNet++



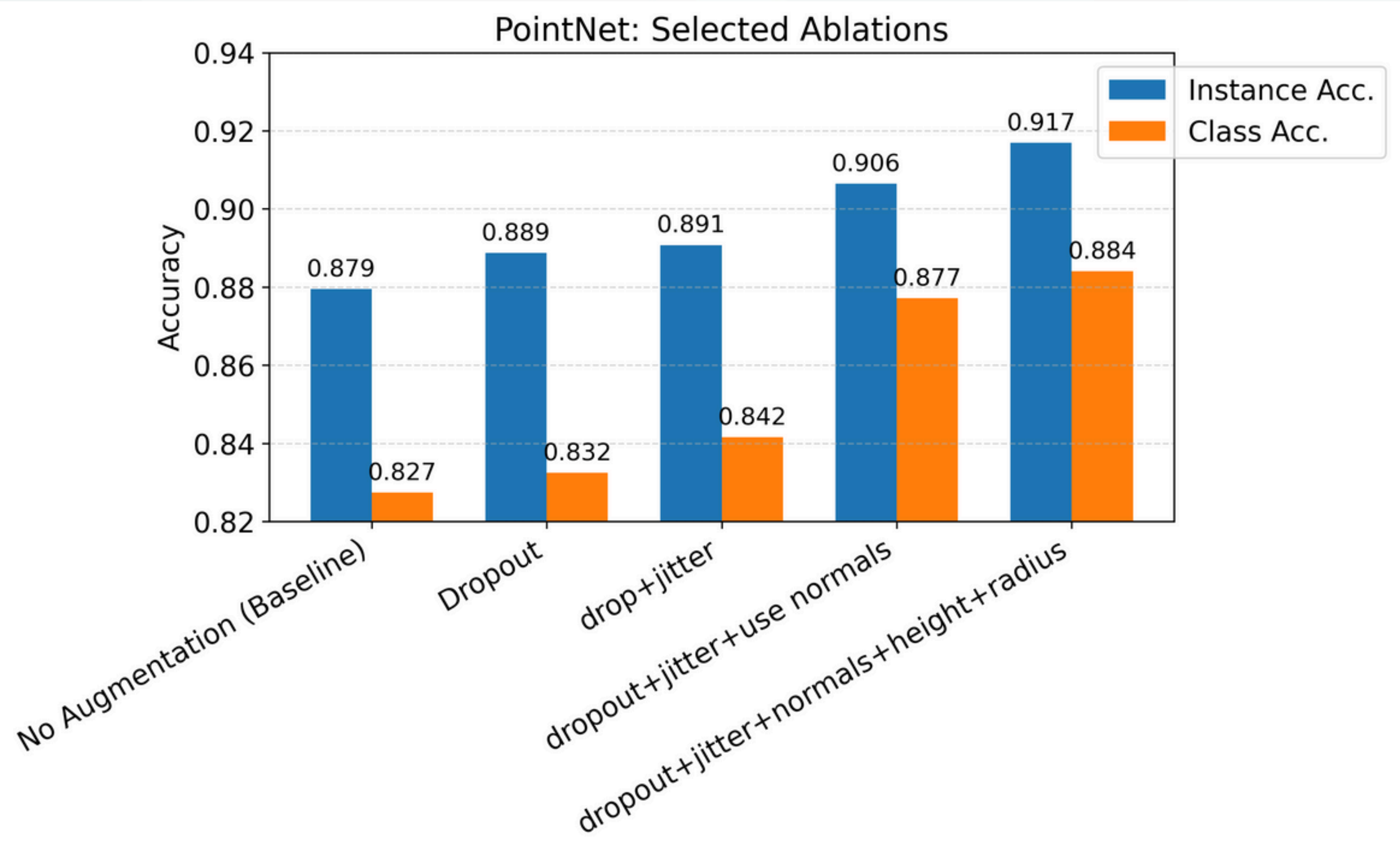
Model Training & Fine-Tuning

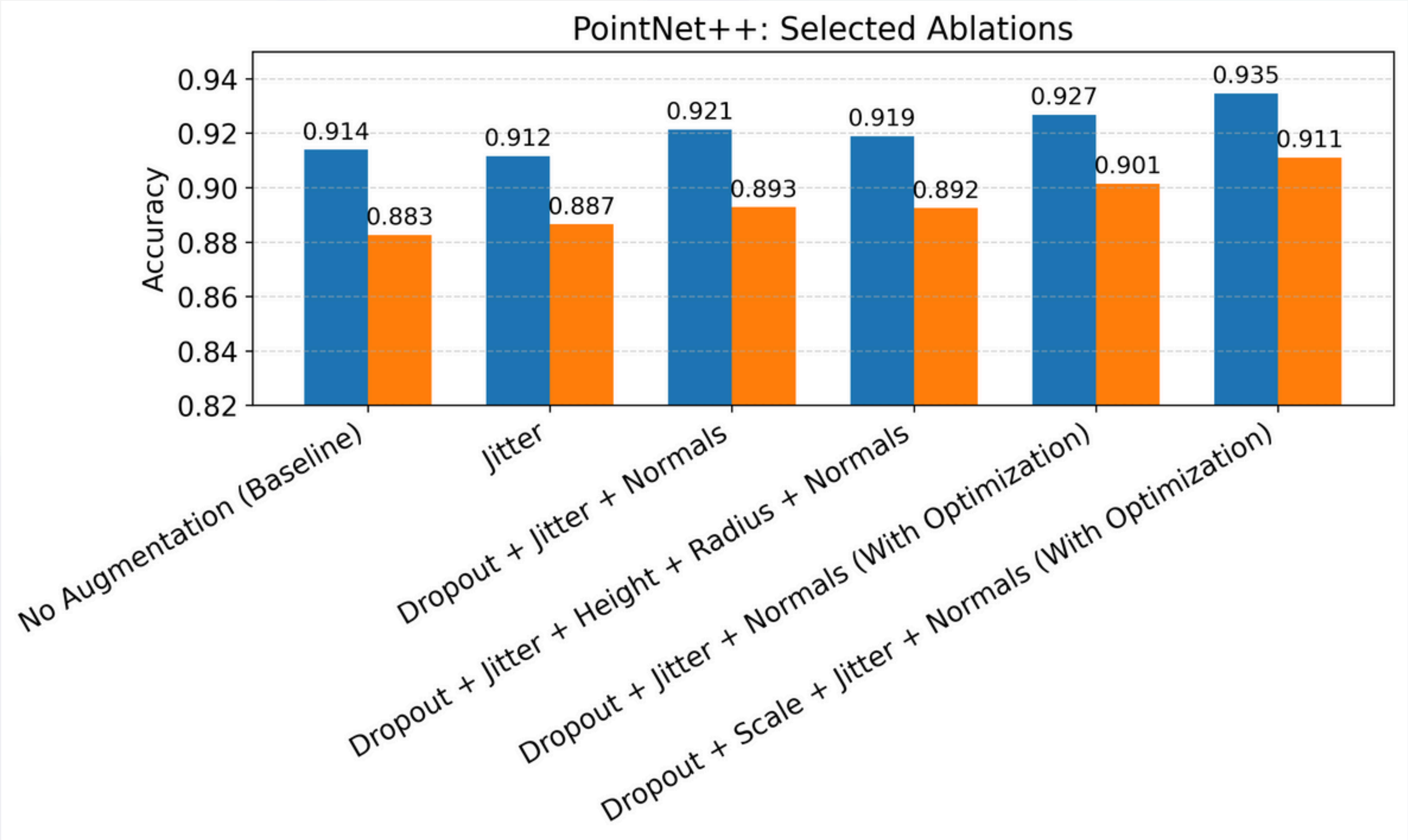
Data Augmentation	Feature Engineering	Optimization Training
Combats overfitting on small datasets (ModelNet40) by synthesizing variations.	Enhances input vectors that feeding into MLP beyond raw XYZ coords.	Refined training loop for stability and convergence
Geometric Transforms: • Rotation (Arbitrary 3D axis) • Scaling • Translation shift • Mirroring **Noise Injection:** • Gaussian jittering with sigma(0.01)& clipping(0.05) • Random point dropout (0 ~ 0.875)	**Extending MLP input point dimension:** (N, 3) → (N, 3 + C) **Explicit features per point:** • Surface Normals (+3) • Local Curvature (+1) • Point Density (+1) • Normalized Height & Radius (+2) • Local PCA components (+3)	**Refinements:** StepLR → Cosine Annealing (to 1e-5) Adam → AdamW (with 0.05 Weight Decay) Standard CE → Label Smoothing (ε = 0.2)

Baseline Models: PointNet & PointeNet++ with Single Scale Grouping (SSG)

Default Training Configuration: --NVIDIA T4 GPU --50 Epochs --LR 0.01 --24 Batch Size

Results & Discussion





PointNet best (Dropout+Jitter+Normals+Height+Radius): - InstAcc 0.879→0.917 (+3.8%) | ClassAcc 0.827→0.884 (+5.7 %)

PointNet++ best (Dropout+Scale+Jitter+Normals with Optimization): InstAcc 0.914→0.935(+2.1%) | ClassAcc 0.883→0.911(+2.8 %)

Takeway

1. PointNet + aug + features → reaches PointNet++ baseline accuracy
2. Rotation hurts (breaks canonical orientation); dropout, jitter, scale help
3. PointNet: normals / height / radius → strong accuracy boost
4. PointNet++: extra features give little gain (SA layers already implicitly capture geometry)
5. PointNet++ benefits more from aug + normals + optimizer/scheduler tuning

Thank You